

Mesith Rathnayake

B.Sc. (Eng) in Computer Engineering (UG)

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👤 Mesith Rathnayake 🔗 Mesith Rathnayake | Portfolio

📍 No 97/A/1/2, Wataraka East, Galle

SUMMARY

Computer Engineering undergraduate at the University of Ruhuna with hands-on experience in software development, Artificial Intelligence, Machine Learning, and Computer Vision. Skilled in Python, image processing, data handling, and developing AI-based applications. Seeking an AI/ML internship to apply technical skills, gain industry experience, and contribute to real-world AI solutions.

SKILLS

- AI & Computer Vision: Python, PyTorch, Scikit-Learn, OpenCV, CNNs, RAG, LLM Integration
- Web & Backend: FastAPI, React.js, Node.js, TypeScript
- Data & MLOps: ChromaDB (Vector), MongoDB, Docker, Jenkins, AWS, Git/Linux

PROJECTS

AI Photo Enhancer – Hybrid Deep Learning & Computer Vision Image Restoration System

Tech Stack: Python, FastAPI, React.js, TypeScript, PyTorch, OpenCV, Real-ESRGAN, GFPGAN, Restormer, Vite, NumPy, Pillow

- **Architected** an end-to-end hybrid image enhancement and restoration pipeline orchestrating deep learning models (**Real-ESRGAN**, **GFPGAN**, and **Restormer**) for blind deblurring, 2x/4x super-resolution, and facial feature reconstruction.
- **Engineered** custom classical computer vision modules with **OpenCV** and **NumPy** for radial lens distortion correction, digital zoom artifact recovery, localized edge-preserving skin smoothing, and color profile preservation.
- **Developed** high-performance asynchronous **FastAPI** backend endpoints with memory-aware tiled GPU inference to eliminate Out-Of-Memory (OOM) errors, supporting single-image stream processing and batch ZIP exports.
- **Built** a responsive, interactive user interface in **React.js**, **TypeScript**, and **Vite**, featuring a real-time before/after image comparison slider and dynamic export controls for sharpening, saturation, and print-ready PPI.
- **Designed** a privacy-first, local-execution architecture ensuring zero external data leakage, backed by robust defensive error boundaries, logging, and high-fidelity JPEG metadata preservation.

My Academia – AI-Powered Academic Tracking System 🔗

Tech Stack: Python, FastAPI, React.js, Node.js, Express.js, MongoDB, ChromaDB, Google Gemini API, Docker, Jenkins

- **Architected an AI-powered academic assistant** using **Retrieval-Augmented Generation (RAG)**, integrating **FastAPI**, **Google Gemini API**, and **Sentence-Transformers** for contextual question answering and automated quiz generation.
- **Engineered an end-to-end document ingestion pipeline** utilizing **PyMuPDF** for text chunking and **ChromaDB** for vector storage and semantic similarity search with source citations.
- **Developed full-stack web features** with React.js, Express.js, and MongoDB, delivering secure user authentication, module management, and academic progress tracking.
- **Containerized microservices** using **Docker** and automated continuous integration/deployment (CI/CD) pipelines with **Jenkins**.
- **Configured and deployed on a self-hosted Ubuntu Linux server** with secure tunneling, ensuring high availability and seamless remote access.

Classroom Engagement Monitoring System 🔗

Tech Stack: Python, Streamlit, OpenCV, dlib, NumPy, SciPy, computer vision, facial landmark detection.

- Built an exploratory computer vision system to analyze student engagement using facial landmark detection techniques.

- Implemented basic attention estimation using Eye Aspect Ratio and head pose concepts with OpenCV and dlib.
- Developed a simple Streamlit interface to visualize live video feed and engagement-related metrics.

BudgetBuddy – DevOps Implementation (Hands-on Learning Project)

Tech Stack: React, Node.js, Express.js, MongoDB, Docker, Docker Compose, Jenkins, Terraform, Ansible, AWS EC2, GitHub, Docker Hub, Grafana, Prometheus

- Designed and deployed a full-stack application to gain hands-on experience with DevOps practices, including CI/CD pipelines, Infrastructure as Code, and production monitoring.
- Implemented a Jenkins CI/CD pipeline triggered by GitHub webhooks to automate build, containerization, and deployment processes to AWS EC2.
- Integrated monitoring and health-check workflows to track application availability and improve incident visibility during deployments.
- Reduced manual deployment effort by automating build and release workflows, improving consistency, reliability, and recovery speed.
- Utilized Docker and Docker Compose for containerization, Terraform for infrastructure provisioning, and monitoring tools to support repeatable and observable deployments.

LEADERSHIP & VOLUNTEERING EXPERIENCE

District Director - Member Relations & Fellowship

07/2025 – Present

Leo District 306 D8 | Lions International

- Oversee member engagement, fellowship activities, and inter-club coordination across the district
- Contribute to strategic planning and leadership development initiatives

Junior Treasurer

02/2026 – Present

REF Media - Faculty of Engineering, University of Ruhuna

- Support financial planning and budgeting for media-related events and productions
- Coordinate videography and live-streaming operations for faculty-level events
- Assist in managing media teams, production workflows, and technical setups for live broadcasts

EDUCATION

B.Sc. (Engineering) [Hons], Computer Engineering

05/2023 – Present

Faculty of Engineering, University of Ruhuna

G.C.E Advanced Level

01/2018 – 08/2021

Richmond College

Physical Science - AAA

G.C.E Ordinary Level

02/2007 – 12/2017

Richmond College

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ACHIEVEMENTS

Second Runners-Up at VoltCast 1.0 Competition

IEEE PES Student Branch and the Electrical & Information Engineering Society, University of Ruhuna (Team ScorpionX).

Secured a Top 8 position at Eminence 5.0

Ranking among the Top 10 teams overall, competing in Power System Simulation (MATLAB), Network Design (Packet Tracer), Machine Learning, and Innovation Project Pitching as part of Team Quadra.